**Latest version**: `https://vuphan314.github.io/`
This version: 2017/08/09

# Vu Phan

*CV*


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vuphan314


## Education

2017/08–present **Doctor of Philosophy**, *Computer Science*.
- Rice University, Houston TX.

2014/08–2017/08 **Dual Bachelor of Science**, *Computer Science & Math*.
- Texas Tech University, Lubbock TX.
- GPA 4/4.

2013/01–2014/08 **Associate of Science**.
- Collin College, Plano TX.
- GPA 4/4.

## Research

### Papers

2016–Present **Syntactic Conditions for Antichain Property in `CR-Prolog`**, *Logic Programming*.
- Discovered syntactic conditions which ensure that a `Consistency Restoring Prolog` program has antichain property (no answer set is a proper subset of another).
- Revising my paper before submitting for publication.

Advisor: Dr. Michael Gelfond (Computer Science Department, Texas Tech University).

2013–Present **Bi-Quasi-Arithmetic Mean Comparability**, *Real Analysis*.
- Generalized a theorem on inequalities involving quasi-arithmetic means.
- Revising my paper before resubmitting to Rose-Hulman Undergraduate Math Journal (previously submitted in 2013 & 2014).
- Presented my paper at the 2013 Texas Undergraduate Math Conference.

Advisor: Dr. Alan Graves (Math Department, Collin College).

### Projects

2017 **Inverse Game of Life**, *Complexity Theory*.
- Investigated the time complexity to invert Conway's Game of Life.

Advisor: Dr. Chris Monico (Math Department, Texas Tech University).

2015 **The Non-Commutative A-Ideal of the Three-Twist Knot**, *Geometric Topology*.
- Did skein computations in the complement of the knot.

Advisor: Dr. Razvan Gelca (Math Department, Texas Tech University).

# Work

## Software Developing

2016–2017 **`LED`**, *Literate Programming System*.

- Implemented the `Language of Effective Definitions` by automating its translation into the functional programming language `SequenceL`.
- Implemented an `LED`-to-`TeX` translator.

Supervisor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University).

2016 **`Knotty`**, *Computer Algebra System*.

- Designed the procedural programming language `Knotty` for knot theory.
- Implemented `Knotty` by automating its symbolic computation and `TeX` printing.
- Implementing additional knot-theoretic operations in `Knotty`.

Supervisor: Dr. Razvan Gelca (Math Department, Texas Tech University).

2015–2016 **`L`**, *Logic Programming Engine*, NASA Research Grant.

- Implemented the language `L` by automating its translation into `Answer Set Prolog`.

Supervisor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University).

## Tutoring

2015 **Computer Science, Math, Physics, Chemistry**.
Supervisor: Mr. Patrick Bohn (Learning Center, Texas Tech University)

2014 **Math, Physics, Chemistry**.
Supervisor: Mr. Dennis Keeton (Math Lab, Collin College)

# Math Contests

2016 **William Lowell Putnam Math Competition**, *Mathematical Association of America*.

- Participated (rank 733/4164, score 20/120).

2013–2014 **Student Math League**, *American Mathematical Association of Two-Year Colleges*.

- Won Southwest Region (AZ, NM, OK, TX)
  in 2012-2013 (round 2) and 2013-2014 (round 2 & aggregate).

2013 **Math Competition**, *Collin College*.

- Won Precalculus Section.

# Scholarships

2016–2017 **General Scholarship**, *Financial Aid Office*, Texas Tech University.

2015–2017 **Academic Achievement Scholarship**, *College of Arts & Sciences*, Texas Tech University.

2015–2017 **Undergraduate Achievement Scholarship**, *Math Department*, Texas Tech University.

2015–2016 **Undergraduate Research Scholarship**, *Math Department*, Texas Tech University.

2014–2016 **Proven Achiever Scholarship**, *Financial Aid Office*, Texas Tech University.

2013–2014 **Excellence Scholarship**, *Math Department*, Collin College.

## Courses

2017 Spring **Abstract Algebra 2**.
Instructor: Dr. Dermot McCarthy (Math Department, Texas Tech University)

2016 Fall **Databases**.
Instructor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University)

2016 Spring **Algorithms**.
Instructor: Dr. Michael Gelfond (Computer Science Department, Texas Tech University)

2015 Fall **Real Analysis 2**.
Instructor: Dr. Razvan Gelca (Math Department, Texas Tech University)

2015 Spring **Automata**.
Instructor: Dr. Michael Gelfond (Computer Science Department, Texas Tech University)

2015 Spring **Programming 1**.
Instructor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University)

2014 Fall **Proofs**.
Instructor: Dr. Chris Monico (Math Department, Texas Tech University)

## Standardized Tests

2015 **GRE (Graduate Record Examination)**, *General Test*.

| section | score | percentile |
|---|---|---|
| Quantitative | 168 / 170 | 95 |
| Verbal | 153 / 170 | 60 |
| Writing | 3.5 / 6 | 42 |

2012 **TOEFL (Test of English as a Foreign Language)**, *Internet-Based Test*.

| Total | 104 / 120 |
|---|---|
| Reading | 29 / 30 |
| Listening | 28 / 30 |
| Writing | 24 / 30 |
| Speaking | 23 / 30 |

## Computer Skills

- Programming Languages:
  - Procedural: `Knotty` (my design & implementation), `Python`/`Java`/`C++`/`C`, `Assembly`
  - Logic: `L` (my implementation), `SPARC`/`Clingo`/`DLV`
  - Literate: `LED` (my implementation)
  - Functional: `SequenceL`, `Scheme`
  - Formatting: `TeX`, `HTML`
  - Data: `SQL`
- Tools: Git/GitHub, Linux/Windows shells